

Education

(Nov 2025-ongoing) Institut national de la recherche scientifique (INRS), Varennes, Québec, Canada

MSc in Energy and Materials (CGPA: 4.3/4.3, up to now)

Thesis: Quantum Advantage on Generative Machine Learning problems on a photonic platform

Supervisor: Prof. Roberto Morandotti

(Feb 2020-March 2025) Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

BSc.Eng. in Electrical and Electronic Engineering (CGPA: 3.9/4.0)

Thesis: Photonic Memory based on Phase Change Materials for Neuromorphic Computing

Supervisor: Prof. Md Zahurul Islam

Research Experience

(September 2025-ongoing) INRS — Master's Student (Supervisor: Prof. Roberto Morandotti)

My work focuses on quantum machine learning, aiming to demonstrate quantum advantage in generative AI tasks on photonic platforms. I am responsible for formulating the research problem, defining realistic physical and computational constraints, and developing a coherent project plan that bridges theory and experiment. In practice, I am building the theoretical and numerical foundations by studying quantum complexity theory and discrete-variable quantum machine learning, and translating these ideas into continuous-variable and high-dimensional (qudit) photonic implementations. This involves analyzing algorithms, identifying feasible photonic architectures, and evaluating trade-offs between theoretical performance and physical realizability.

(May 2024–April 2025) North South University (NSU) Optics Lab — Research Assistant (Supervisor: Prof. Mahdy Rahman Chowdhury)

During my undergraduate studies at BUET, I worked at the NSU Optics Lab on near-field energy transfer between a thermal emitter and an InSb TPV cell, studying how extreme near-field coupling enhances spectral selectivity and improves TPV efficiency beyond the blackbody limit. I designed and simulated a metamaterial perfect-absorber thermal emitter matched to the TPV band edge and analyzed the roles of evanescent modes, surface polaritons, and emitter–cell separation. This work resulted in a journal paper currently awaiting publication at the *Journal of Applied Physics*.

(Feb 2024–March 2025) BUET — Undergraduate Researcher (Supervisor: Prof. Md. Zahurul Islam)

For my undergraduate thesis at BUET, I developed an all-optical memory cell based on GSST ($\text{Ge}_2\text{Sb}_2\text{Se}_4\text{Te}_1$) for photonic neuromorphic computing, resulting in a publication in *Optics Express*. The work involved modeling GSST switching behaviour, optimizing device structures, and evaluating performance for neuromorphic photonic circuits. I also contributed to a number of related projects, including deep-learning–based inverse design of phase-change memory, numerical modeling of multi-level memory behavior, and system-level simulations of neuromorphic photonic architectures using phase-change materials as programmable weights. Collectively, these projects strengthened my expertise in photonic memory design and computational modeling.

(July 2021–August 2023) Team Interplanetar, BUET Mars Rover Team, Bangladesh — Electrical and Software Project Manager (Supervisor: Prof. Kazi Arafat Rahman)

After being selected through competitive exams in 2021, I contributed to Team Interplanetar by designing the Power Distribution Board PCB for the Procheshta v1.0 rover, capable of safely handling over 100 A of DC under stall conditions for reliable field operation. I later represented the team at the University Rover Challenge (URC) 2023 at the Mars Desert Research Station in Utah, USA, participating on a scholarship from Grameenphone, Bangladesh's largest telecom company.

Peer-Reviewed Journal Articles

1. **Chinmoy Biswas**, Jannatul Ferdous, Zahin Ikbāl Zeal, AKM Hasibul Hoque, Ying Yin Tsui, and Md Zahurul Islam. "High-contrast and stable operation of a plasmonically-enhanced GSST-based phase-change memory cell using plasmonic chain modes for photonic memory applications." *Opt. Express* **33**(20), 43393-43403 (2025).
2. **Chinmoy Biswas**, Habiba Akter, Farhana Islam, Mahdy Rahman Chowdhury, "Harnessing Near-Field Interactions in a Perfectly-Tuned Metamaterial Selective Emitter for High-Efficiency InSb Thermophotovoltaics", *Journal of Applied Physics* **139**(7), accepted on 02/12/2025 and awaiting publication (Editor's Pick).

Submitted Journal Articles

1. **Chinmoy Biswas**, Nafis Faisal, Vivek Chowdhury, Abrar Al-Shadid Abir, Sabir Mahmud, Mithon Rahman, Hafiz Imtiaz, Shaikh Anowarul Fattah, "Load forecasting on a highly sparse electrical load dataset using Gaussian interpolation", currently under review in *Journal of Forecasting*.

Supervisory and Mentorship Experience

At Team Interplanetar, I led a practical course on Arduino and ROS, leading students through robotics fundamentals and actual project workflows. I also acted as an academic mentor for 11th and 12th grade high school students, providing guidance in physics and mathematics. At Team Interplanetar, I led a 12-member group responsible for developing electrical components and inverse-kinematics software for our URC 2023 rover.

In addition, I served as a research mentor for four undergraduate students under my supervisor at BUET, guiding them through their research projects, design, documentation, and technical skill development. The students and their thesis topics are listed below:

(March 2025-ongoing) Md. Nagib Mahfuz, BUET, Thesis: "*Deep Learning based Inverse Design of Plasmon Enhanced Photonic Phase Change Memory*".

(March 2025-ongoing) Humayun Ahmed, BUET, Thesis: "*Plasmon Enhanced Phase Change Memory on a Photonic Crystal Platform*".

(March 2025-ongoing) Meftaul Alam Seyam, BUET, Thesis: "*Multi-level storage dynamics of a GSST based photonic phase change memory*".

(May 2025-ongoing) Jannatul Ferdous, BUET, Thesis: "*Multi-shot laser ablation of Silicon waveguides under high fluence femto-second laser irradiation*".

Industrial Experience

- **Industrial Trainee, Neural Semiconductor (2024)**

During my undergraduate studies, I completed an industrial attachment focused on VLSI for over a month, where I learned industrial-scale IC design and manufacturing.

Competitions, Awards, and Fellowships

At BUET during Bachelor's (2020-2025)

- **2nd Position, ULKASEMI VLSITHON 2.0, RTL Section (2025)**
- Three-member team, national Ulkasemi VLSI competition (30+ teams) in East West University; designed an IC from specs in a timed exam, won 50,000 BDT (555 CAD) prize.
- **3rd Position, ULKSEMI VLSITHON 2.0. Analog Section (2025)**
- Ulkasemi VLSITHON 2.0 (Analog), 20+ teams in University of Asia Pacific; designed analog ICs in a timed challenge, won 25,000 BDT (278 CAD) prize.
- **3rd Position, BUET EEE DAY 2024 VLSITHON (2025)**
During the bi-annual event of the EEE department of BUET (EEE DAY), a team VLSI competition was arranged by Neural Semiconductor. The competition was attended by over 20 teams from across Bangladesh at BUET, where we solved IC-related problems in both RTL and Analog domains. I won a cash prize of 10000 BDT (111 CAD).
- **RISE Undergraduate Research Grant (2025)**
The grant was awarded for an innovative undergraduate research thesis (*Nanophotonic Memory Cell Improved by Plasmonic Chain Modes*): a total of 10000 BDT (111 CAD)
- **Dean's List Award (2020-2025)**
In each year of my undergrad, I received the Dean's list award for achieving a CGPA of above 3.75 in one academic year, amassing over 20000 BDT (222 CAD)
- **University Merit Scholarship (2020-2025)**
In each semester of my undergrad, I received the University Merit scholarship for achieving in the top 10% in that semester, amassing over 25000 BDT (278 CAD)
- **Scholarship for work on Team Interplanetar (2023)**
This scholarship was issued by Grammenphone for traveling to the USA for URC2023, totaling over 113000 BDT (1252 CAD)
- **Government Merit Scholarship – Bangladesh (2012–2025)**
Granted by the Government of Bangladesh to students who achieved top ranks in both the Primary School Certificate (PSC, Grade 5), Junior School Certificate (JSC, Grade 8), Secondary School Certificate (SSC, Grade 10), and Higher Secondary Certificate (HSC, Grade 12) national examinations. The scholarship included a \$800 stipend and was awarded in recognition of sustained academic excellence over multiple years.

Voluntary Academic Services

- Collaborated with BUET faculty to organize International Conference on Telecommunications and Photonics (ICTP) 2023 and the International Conference on Electrical and Computer Engineering (ICECE) 2024; served as graphic designer for conference materials including banners, covers, invitations, and presentation templates.

Professional membership

- Student Member, SPIE – The International Society for Optics and Photonics (2025)
- Student Member, IEEE – Institute of Electrical and Electronics Engineers (2019–2025)